FINAL YEAR PROJECT PROPOSAL

# DEEPFAKE VIDEO DETECTION AND CONTENT VERIFICATION SYSTEM FOR MEDIA FORENSIC ANALYSIS


**IQRA NATIONAL UNIVERSITY, PESHAWAR**

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*A project proposal submitted to The IQRA National University, Peshawar in partial fulfillment of the requirements for the degree of Bachelor of Science (BS) in Computer Science.*


**SCHOOL OF COMPUTER SCIENCE**

**IQRA NATIONAL UNIVERSITY, PESHAWAR-PAKISTAN**


**December, 2025**

## 1. Introduction & Problem Statement

The research framework titled "Deepfake Video Detection and Content Verification System for Media Forensic Analysis" targets vital architectural paradigms within the contemporary computer science landscape. Because modern information engineering systems are moving rapidly toward decentralized, agile spaces, specialized automated monitoring frameworks alongside intelligent optimization pipelines become absolutely paramount to unlock next-generation capabilities.

> **Problem Statement:** The aggressive evolution of hyper-realistic generative AI models empowers malicious individuals to weaponize synthetic deepfake video clips. This paradigm rapidly destabilizes socio-political systems, manipulates financial market conditions, and fabricates highly damaging, defamatory narratives without robust verification mechanisms standing in the way. Without an automated, dependable system architecture designed to simplify these computational workflows, operational risks continue to escalate drastically. This platform fills this missing gap by constructing an optimized algorithmic infrastructure that serves as a resilient mechanism for resolving these complex data integrity challenges.

## 2. Project Objectives

The core structural targets and milestones mapped for this software engineering lifecycle include:

- Designing a highly efficient spatiotemporal CNN-LSTM architecture that tracks microscopic inter-frame facial discrepancies and subtle blinking frequencies.
- Constructing a rapid verification extraction pipeline capable of automatically reviewing audio-to-visual voice lip sync alterations.
- Deploying an unalterable, cryptographic blockchain indexing registry that saves valid media file structural hash keys securely.

## 3. Methodology & System Architecture

The software infrastructure strictly adheres to a highly structured, modular processing layout constructed on a decoupled Client-Server model. This design paradigm guarantees that intensive backend model inference pipelines do not disrupt or degrade the front-end user interface application execution parameters.

### Primary Engineering Modules:

- **Authentication & Profile Integration Module:** Handles user record enrollment sequences and secures background data caching streams smoothly.
- **Core Execution and Inference Pipeline:** Supervises live data intake pipelines, transformation tasks, and algorithmic processing nodes.

- **Analytics & Reporting Layout Engine:** Gathers raw calculation metrics to compile structured system diagnostics reports safely.

### System Architecture Blueprint:

> ***React Native Client Interface → Secure Gateway APIs → Node Compute Processing Engines → Scalable Vector/ Relational Databases***

## 4. Proposed Time Schedule

The structural implementation strategy follows a strict timeline sequence spread across the academic year:

| Phase | Description of Activity | Duration (2025-2026) |
|---|---|---|
| **Phase 1** | Literature Review & Core Component Selection | December |
| **Phase 2** | Architecture Design & Structural Integrity Verifications | January – February |
| **Phase 3** | Main Engine Development & Coding Implementation Cycle | March – April |
| **Phase 4** | System Integration Testing & Proposal Synthesis | May – June |

## 5. Budget Description & Technical Stack

The software application fully utilizes open-source libraries alongside public cloud deployment environments, hitting maximum performance efficiency with negligible capital outlays. All heavy computational processing workloads and deep learning model training sessions are run directly on standard developer workstation hardware platforms already accessible to the core engineering team.

> ***Proposed Technical Core Stack:*** *React Native Mobile View, Python Django Framework, OpenCV, PyTorch, Librosa Feature Engine, and Ethereum Testnet architecture.*

## 6. Undertaking & Project Declarations

We explicitly declare that this technical engineering proposal represents our authentic development efforts completed under guidance at IQRA National University. No segment of this engineering research design has been previously handed in for academic credit at any alternative educational institution or board.

Student 1 Signature

Student 2 Signature